

DONGFANG, China

WMO: 598380

Lat: 19.0941N

Lon: 108.6216E

Elev: 26

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 54.0	(c) 56.2	(d) 43.3	(e) 41.4	(f) 58.9	(g) 46.8	(h) 47.3	(i) 59.8	(j) 25.1	(k) 58.9	(l) 23.1	(m) 59.4	(n) 14.8	(o) 40	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.0	(c) 91.9	(d) 81.1	(e) 91.0	(f) 80.7	(g) 90.3	(h) 80.4	(i) 82.8	(j) 89.6	(k) 82.2	(l) 89.1	(m) 81.6	(n) 88.5	(o) 14.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.9	160.5	87.8	80.1	156.5	87.3	79.7	154.1	86.9	46.8	90.1	46.1	89.3	45.5	88.6	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.2	(b) 21.9	(c) 20.0	(e) 51.5	(f) 93.9	(g) 2.6	(h) 1.9	(i) 49.6	(j) 95.2	(k) 48.1	(l) 96.3	(m) 46.6	(n) 97.4	(o) 44.7	(p) 98.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			46.6	83.2	3.1	1.1	44.4	83.9	42.6	84.6	40.9	85.2	38.6	85.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.5	67.7	69.3	74.0	80.6	84.4	86.1	85.3	84.4	82.7	80.4	76.0	70.1
	DBStd	7.56	5.23	6.24	5.38	4.25	3.04	2.30	2.36	2.08	2.23	2.62	4.09	4.93
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	93	37	32	7	0	0	0	0	0	0	0	1	16
	CDD50	10386	548	540	743	917	1067	1083	1093	1067	982	943	780	623
	CDD65	5005	121	151	285	468	602	633	628	602	532	478	331	174
	CDH74	52976	310	599	1751	4867	7731	8633	8258	7586	6003	4471	2161	606
	CDH80	20818	9	51	309	1673	3534	4357	3872	3226	2090	1243	411	43
(14) Wind	WSAvg	9.5	9.3	8.8	8.4	9.0	9.8	11.6	10.9	9.0	8.0	8.5	9.9	10.3
(15) Precipitation	PrecAvg	36.8	0.3	0.5	0.7	1.1	2.6	5.7	4.3	7.9	6.3	5.2	1.0	0.2
	PrecMax	60.2	1.6	3.7	2.8	3.3	9.9	21.2	17.8	20.4	17.1	15.7	4.8	0.8
	PrecMin	10.8	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.5	0.4	0.2	0.0	0.0
	PrecStd	12.5	0.4	0.7	0.7	1.0	2.0	4.6	5.0	5.4	4.2	3.5	1.0	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.0	83.8	87.1	91.2	93.1	93.1	92.5	92.1	90.7	89.8	87.1	83.3
		MCWB	71.8	74.1	77.0	79.7	81.3	82.3	81.4	81.7	80.2	76.6	75.9	73.9
	2%	DB	78.9	81.6	85.2	89.7	91.6	91.7	91.3	90.5	89.5	88.1	85.3	81.2
		MCWB	70.8	73.2	75.9	79.1	80.9	81.4	80.9	81.1	80.0	77.0	75.7	72.6
	5%	DB	77.3	79.8	83.4	88.4	90.6	90.8	90.3	89.5	88.5	87.0	84.0	79.3
		MCWB	70.0	72.1	74.8	78.4	80.3	80.9	80.5	80.7	79.7	76.9	75.1	71.5
	10%	DB	75.6	77.9	81.5	87.1	89.7	90.0	89.3	88.5	87.5	85.8	82.6	77.6
		MCWB	69.0	71.1	73.8	77.7	79.8	80.5	80.1	80.3	79.3	76.7	74.2	70.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.7	75.5	78.0	81.5	83.4	83.6	83.1	83.2	82.6	80.5	78.8	76.3
		MCDB	79.1	82.2	85.4	89.1	90.4	90.8	89.9	89.6	88.5	86.9	84.3	81.1
	2%	WB	72.0	73.8	76.6	80.3	82.2	82.7	82.2	82.4	81.6	79.4	77.6	74.6
		MCDB	77.0	80.2	83.6	87.8	89.9	90.0	89.1	88.5	87.5	85.5	83.0	79.4
	5%	WB	71.0	72.8	75.5	79.2	81.4	82.0	81.6	81.7	80.8	78.7	76.6	73.2
		MCDB	75.9	78.7	82.1	86.6	89.2	89.3	88.4	87.8	86.7	84.5	81.9	77.7
	10%	WB	69.9	71.6	74.5	78.4	80.6	81.4	80.9	81.0	80.1	78.0	75.7	71.7
		MCDB	74.6	76.9	80.6	85.6	88.4	88.7	87.7	87.0	85.8	83.5	80.8	76.2
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	9.5	9.3	9.0	8.4	7.0	7.0	7.6	8.6	10.0	10.4	10.4
		MCDBR	12.1	10.9	10.3	9.5	8.6	7.4	8.0	8.6	9.8	11.6	11.7	12.0
		MCWBR	6.3	5.9	6.0	5.9	5.2	4.7	4.6	4.1	4.3	4.6	5.3	5.9
	5% WB	MCDBR	10.7	10.1	10.0	9.0	8.4	7.2	7.4	7.7	8.9	9.9	9.8	10.4
		MCWBR	6.1	6.0	6.1	5.7	5.5	4.9	4.5	4.2	4.5	4.5	4.9	5.8
(40) Clear-Sky Solar Irradiance	taub		0.556	0.586	0.702	0.656	0.510	0.482	0.462	0.499	0.557	0.615	0.551	0.572
	taud		1.872	1.834	1.630	1.771	2.207	2.300	2.349	2.224	2.017	1.822	1.955	1.851
	Ebn at Noon		228	231	210	222	254	259	265	257	241	220	228	218
	Edh at Noon		60	66	83	73	47	42	40	46	56	66	55	60
(44) All-Sky Solar Radiation	RadAvg		1036	1225	1360	1660	1852	1800	1759	1658	1544	1434	1223	997
	RadStd		165	200	143	149	141	137	178	110	152	148	106	116

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+135
Regional (7 neighbors)	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page